SECTIONS OVER A WEAKLY CONTRACTIBLE OBJECT ARE EXACT, ACYCLIC AND HIT TORSORS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the two statements read off the definition and the vanishing of the higher derived functors of the exact sections functor.

Editorial note. The label lemma-w-contractible names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The statement and proof are copied from the pinned chapter file; the two cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Weakly contractible objects of a site. Same chapter, the definition of a weakly contractible object through the exactness of the sections functor and the triviality of the torsors over it. Supplies the object whose three properties are asserted.
- Right derived functors of an exact functor vanish. Derived Categories, the lemma stating that the right derived functors of an exact functor vanish in positive degrees. Supplies the vanishing of the higher cohomology groups in part (2).

Source reference. Original lemma 015F: sources/stacks-chow/tags.txt, source lines 1506–1506. Prerequisite pointer invoked at proof line(s) 13790; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 010M: sources/stacks-chow/tags.txt, source lines 1332–1332. Prerequisite pointer invoked at proof line(s) 13788; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $\mathcal{C}$ be a site. Let U be a weakly contractible object of $\mathcal{C}$. Then*

(1) *the functor $\mathcal{F} \mapsto \mathcal{F}(U)$ is an exact functor $Ab(\mathcal{C}) \to Ab$,*
(2) *$H^p(U, \mathcal{F}) = 0$ for every abelian sheaf $\mathcal{F}$ and all $p \geq 1$, and*
(3) *for any sheaf of groups $\mathcal{G}$ any $\mathcal{G}$-torsor has a section over U.*

Proof. The first statement follows immediately from the definition (see also Homology, Section 010M). The higher derived functors vanish by Derived Categories, Lemma 015F. Let $\mathcal{F}$ be a $\mathcal{G}$-torsor. Then $\mathcal{F} \to *$ is a surjective map of sheaves. Hence (3) follows from the definition as well. □

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